Chest Patch

Design specification v2.0

Housing lengthened to **110 mm**. The overall size, skin interface, layer stack and rigid-flex PCB envelope have been re-derived from the "Classic Infinity" concept render and fitted to Indian adult chest and sternum data.



110 × 52
HOUSING L × W (MM)

7.3 mm
ON-SKIN HEIGHT, INCL. GEL

58 mm
ECG ELECTRODE SPACING

2,680 mm²
RIGID PCB AREA (3 ISLANDS)

Reference render: concept board "Bionic Chest Patch"

01 SIZE DECISION & BODY PLACEMENT

Is a 110 mm housing OK? Yes, with three conditions.

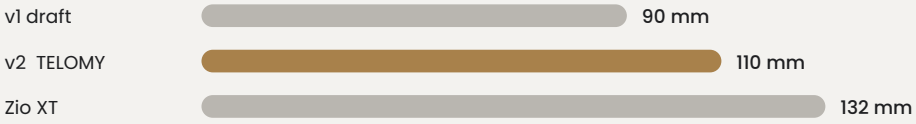
110 mm sits between the first draft (90 mm) and the iRhythm Zio XT (132 mm), which is worn for up to 14 days across the upper-left chest, so this length is proven territory for an adhesive chest patch. It gives you a wider ECG baseline, about 55% more PCB area and a larger battery. The risk is fit on slim or small-framed users, and it is managed by the three conditions below.

	What 110 mm gives you	What to watch
ECG	Electrode spacing 52 to **58 mm** (research: ≥ 50 mm gives a reliable single-lead signal)	Horizontal orientation still gives a smaller R-wave than the diagonal
PCB	Rigid area ≈ 1,730 to **≈ 2,680 mm²**	More copper = more stiffness; keep the waist flexible
Battery	Custom D-shaped pouch ≈ **200 mAh** (was ≈ 120)	Custom cell = MOQ and lead time
Fit	Ends at ±55 mm from midline, adhesive at ±59 mm	Slim chest (R ≈ 140 mm) curves away ≈ 11 mm over 110 mm
Comfort	Weight target ≤ 18 g	Skin stretch when raising arms pulls at the lobe ends

CONDITIONS FOR 110 MM

1. Rigid-flex, not rigid. Two rigid lobes + a flexible waist that bends ±13° per side, over-moulded in soft silicone/TPU.
2. Tapered 4 mm flange. Edge thickness 0.8 mm rising to 1.5 mm, with the adhesive running 4 mm past the housing, so the ends do not peel.
3. High, central placement. Centre it on the sternal angle, no lower. For small-framed and female users, plan a 90 mm "Compact" variant later.

LENGTH COMPARISON (TO SCALE, 1:1.6)

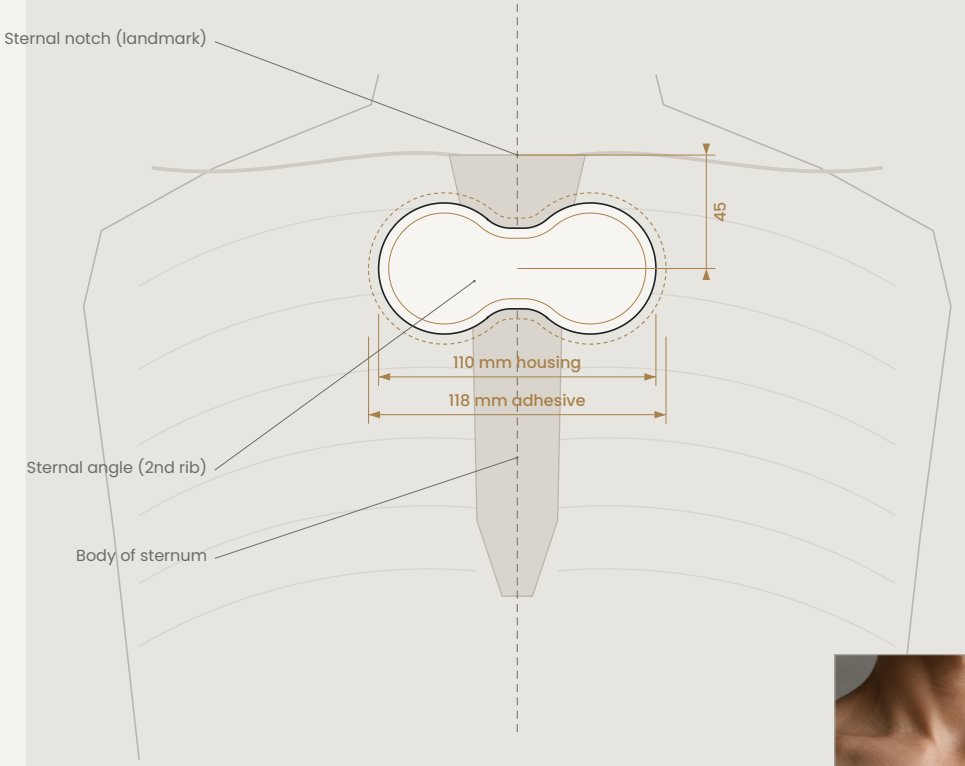


PRIMARY PLACEMENT

Horizontal on the midline, with the centre on the sternal angle (≈ 45 mm below the notch). Top edge at least 15 mm below the notch.

CARDIAC MODE

Rotate 30–45° down-left along the left sternal border (Zio-style, modified Lead II) for a stronger R-wave.



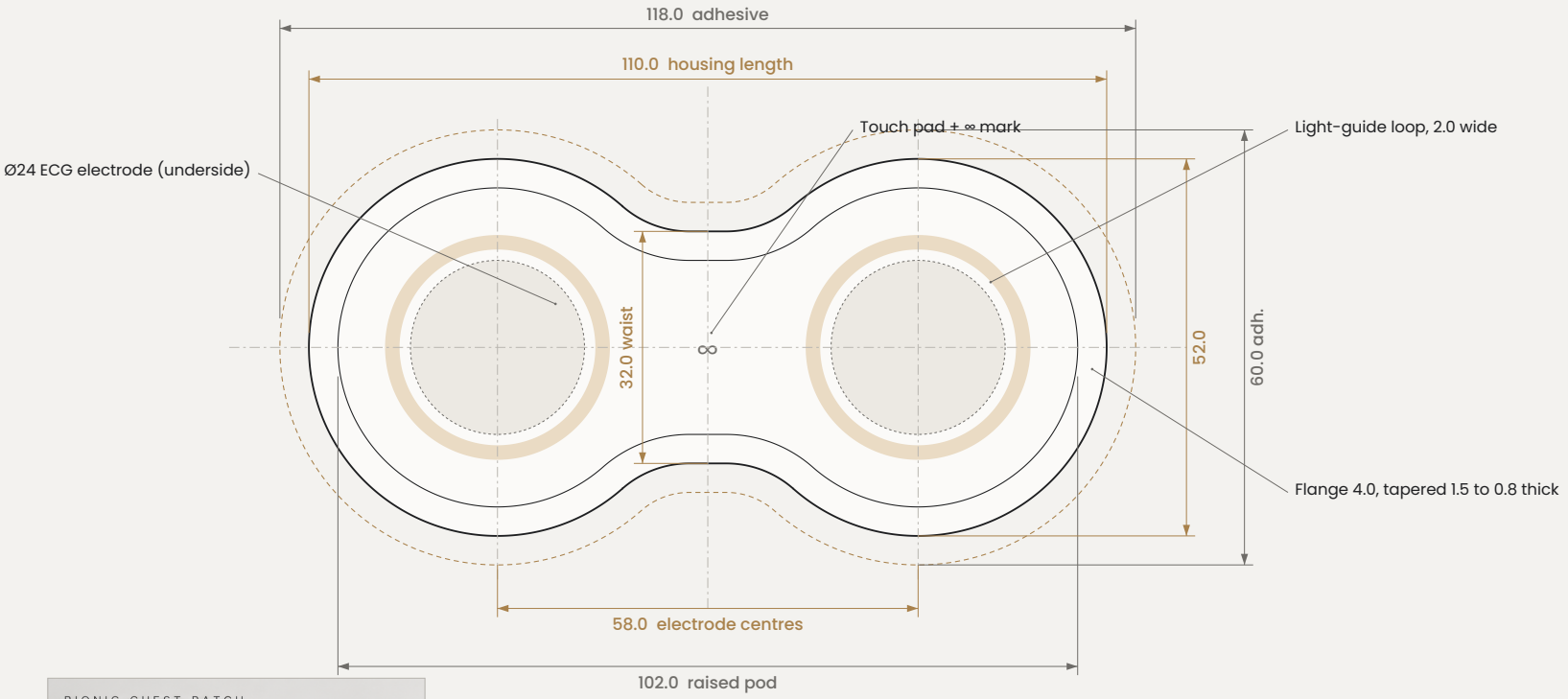
Schematic, 1:3 scale. Slim Indian adult male (chest ≈ 890 mm)

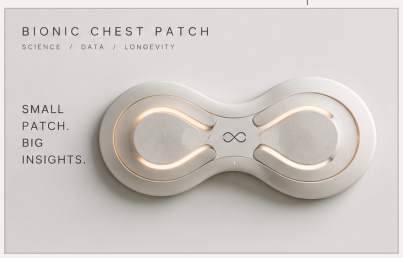


CONCEPT RENDER, ON BODY

02 HOUSING OUTLINE

Top view: overall dimensions (1:1 scale)





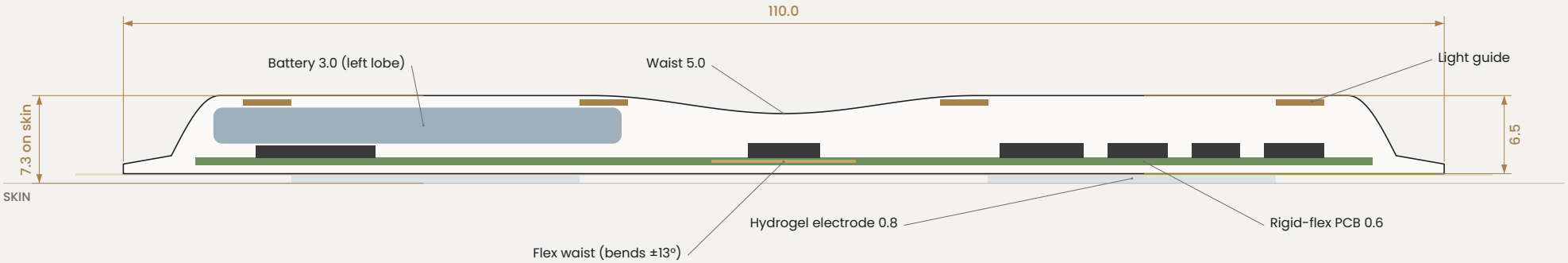
REFERENCE: CLASSIC INFINITY RENDER

SCALE CHECK: THIS BAR = 50 MM WHEN PRINTED AT 100%

Parameter	Value
Housing L × W	110.0 × 52.0
Lobe radius / centres	R26.0 @ ±29.0
Waist height	32.0 (R14 concave blend)
Adhesive offset	+4.0 all round to 118 × 60
Housing footprint	4570 mm²
Adhesive footprint	5770 mm²

03 THICKNESS

Section and layer stack: 6.5 mm housing, 7.3 mm on skin



LONGITUDINAL SECTION THROUGH ELECTRODE CENTRES, 2:1 SCALE

Layer (skin to top)	Thickness	Notes
Hydrogel electrode + adhesive	0.8	Replaceable skin-contact layer; Ag/AgCl on hydrogel
Bottom shell	0.7	PC/TPU; electrode snaps, PPG & temp windows
Rigid-flex PCB (rigid islands)	0.6	4-layer rigid, 2-layer polyimide flex
Components (max height)	1.2	≤ 1.0 under the battery
Li-Po battery (left lobe)	3.0	Custom D-shape pouch ≈ 200 mAh
Clearance / foam	0.2	Swelling allowance
Top shell + light guide	0.8	Biocompatible, soft-touch
Total on skin	**7.3**	Housing only 6.5; waist 5.0; flange edge 0.8



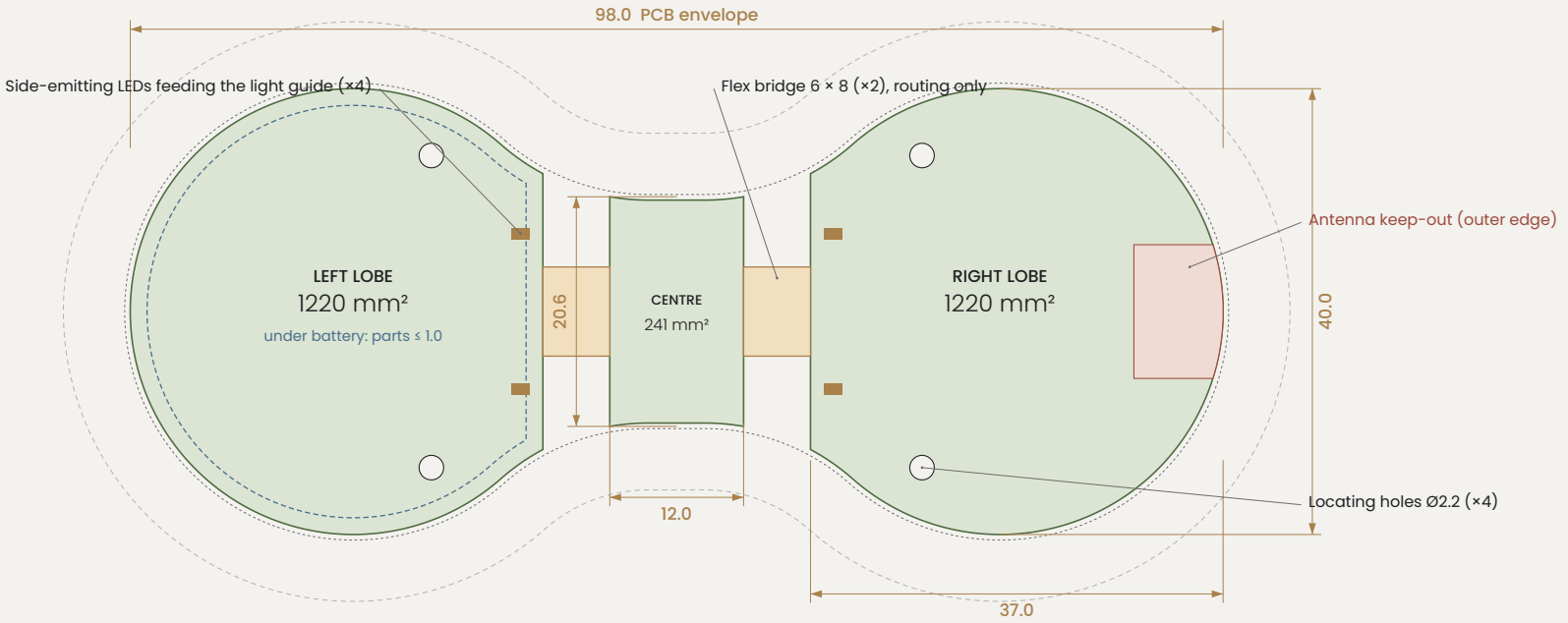




REFERENCE: EXPLODED LAYER CONCEPT

04 PCB DESIGN AREA

PCB envelope: 3 rigid islands on a rigid-flex board



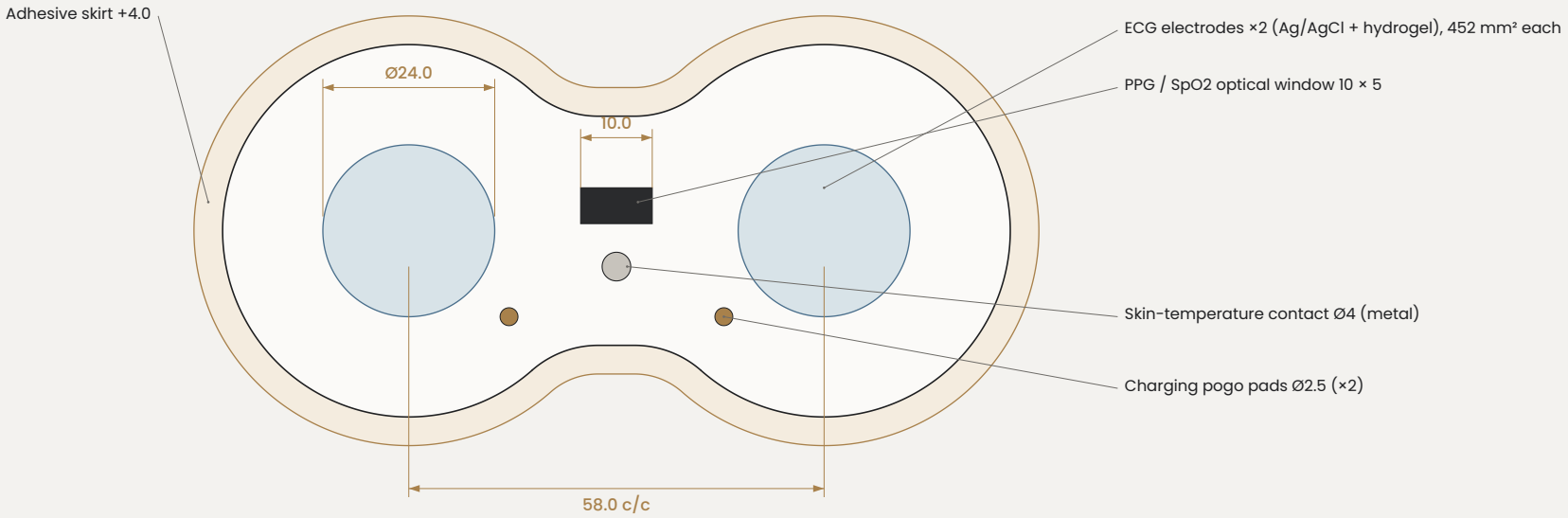
DRAWN AT 1.5:1. PCB envelope = housing outline offset 6.0 inward (1.2 wall + 4.0 flange step + 0.3 draft + 0.5 clearance).

Zone	Area	Suggested contents
Right lobe (top)	1220	BLE SoC (nRF5340 class) + 32 MHz xtal, DC-DC inductor, ECG AFE (MAX30003 / ADS1292R class), 6-axis IMU, 64 Mb flash, chip antenna, SWD pads
Left lobe (top)	1220	PMIC / charger + fuel gauge, battery tabs, ESD, LED driver. Parts ≤ 1.0 mm (battery above)
Centre island	241	Top: capacitive touch pad under ∞, status LED. Bottom: PPG optics (LEDs + 2 PD) and AFE (MAX86141 class), skin-temp sensor on thermal pad
Flex bridges	96	Routing only, no parts, no vias in the bend zone
Total rigid	**2680 mm²**	**≈ 2360 mm² usable** on the top side after keep-outs; bottom side reserved for sensors & contacts

- Rigid island (4-layer, 0.6)
- Flex (2-layer polyimide)
- Battery footprint 1040 mm²
- RF keep-out
- Housing cavity

05 ELECTRODES, SENSORS & ADHESIVE

Underside: skin interface (1:1 scale)



SKIN-SIDE RULES

- Die-cut the adhesive with openings for both electrodes, the PPG window, the temperature pad and the pogo pads.
- Keep the PPG window at least 10 mm from the electrode gel edge so gel does not smear onto the optics.
- Recommended adhesive: breathable medical-grade acrylic or silicone tape. Design for 3 to 5 days of wear per patch in Indian summer; replace the adhesive and gel together.
- Hydrogel wet area per electrode is 452 mm². Do not go below Ø18 or skin-electrode impedance rises.
- Put a thermal barrier (foam) between the battery and the skin-temp sensor so the reading is not biased.



REFERENCE: UNDERSIDE CONCEPT



SENSING STACK: ECG · TEMP · SPO2 · BLE

06 CMF & VARIANTS

Form language and colour, material & finish



CONCEPT VARIANTS. "CLASSIC INFINITY" IS THE BASE FOR THIS 110 MM SPEC

Finish	Approx. hex	Shell material / process	Glow
Pearl White	#E8E4DE	PC/ABS medical grade, soft-touch PU coat	Warm 2700 K
Midnight Grey	#3A3C3E	PC/ABS, fine texture	Cool 6500 K
Champagne Gold	#C9AE92	PC + metallic pigment or PVD ring	Warm 2700 K
Matte Black	#1A1A18	PC, VDI-24 texture	Warm 3000 K
Forest Green	#5E6B61	PC/ABS, soft-touch coat	Neutral 4000 K

NOTES

- The shell touches skin at the flange: use an ISO 10993-5/-10 tested material; waist over-mould in medical LSR (Shore 40–50 A).
- Light guide: 2.0 mm wide PMMA/PC ring following each lobe. Default brightness low; use the glow only for status and session prompts, which also saves battery.
- Hex values are sampled from the render and are a starting point only. Confirm with physical colour chips before tooling.
- The ∞ mark is debossed 0.2 mm and doubles as the capacitive touch zone.



COLOUR OPTIONS FROM CONCEPT BOARD

07 BASIS OF DESIGN

Fit data, checks and sources

Input	Value	Used for
Chest circumference, Indian male (Odisha study)	Mean 886 mm; P5–P95 744–1003	Worst-case curvature (P5)
Inter-scye (shoulder) breadth, Indian male	Mean 336 mm; P5 289 mm	Available flat width on upper chest
Sternum length, North Indian male / female	150–170 / 116–149 mm	Vertical limit (60 mm adhesive fits)
Single-lead ECG electrode distance	≈ 50 mm reliable; best near V1–V4, diagonal	Electrode spacing 58 mm
Zio XT (reference patch)	132 × 51 × 14 mm, 24.5 g	Size benchmark

CURVATURE CHECK (SLIM CHEST, R ≈ 140 MM)

Span	Gap if rigid (L^2 / 8R)	Handled by
Full housing, 110	≈ 10.8 mm	Flex waist: each lobe rotates ≈ 13°
One rigid lobe island, 37	≈ 1.2 mm	0.8 hydrogel + adhesive compliance; pre-curve lobe base R180
Flange edge	n/a	0.8 mm tapered edge follows the skin

OPEN ITEMS BEFORE TOOLING

1. 3D-print 110 mm dummies (rigid lobes + TPU waist) and test-fit on 10–15 people, including slim men and small-framed women.
2. Record ECG with horizontal vs 30–45° placement on the same subjects; confirm R-wave amplitude at 58 mm spacing.
3. Reliable Indian female chest-shape data was not found. Size the 90 mm Compact variant from the fit trials.
4. Validate sternal reflectance SpO2 early. Perfusion there is lower than at the wrist or forehead.
5. Battery-life figure is an estimate (≈ 1.2 mA average to about a week). Measure it on the EVT board.
6. ECG / SpO2 claims make this a medical device: plan the CDSCO pathway, IEC 60601-1 / -1-2 and ISO 10993.

SOURCES

1. Anthropometric dimensions among Indian males: a principal component analysis (dergipark.org.tr/en/download/article-file/808374)

2. Gender determination from sternum in living North Indian adults, IJASHNB (ijashnb.org/archive/volume/4/issue/4/article/6079)

3. Sexual dimorphism of the human sternum, Maharashtrian population (researchgate.net/publication/6142030)

4. Optimized Electrode Locations for Wearable Single-Lead ECG Monitoring, Sensors 2019 19(20):4458 (mdpi.com/1424-8220/19/20/4458)

5. Optimal Lead Position in Patch-Type Monitoring Sensors, Sensors 2020 (pmc.ncbi.nlm.nih.gov/articles/PMC7071434)

6. iRhythm Zio XT Clinical Reference Manual (irhythmtech.com)

7. Concept board "Bionic Chest Patch" (supplied render; proportions only)



